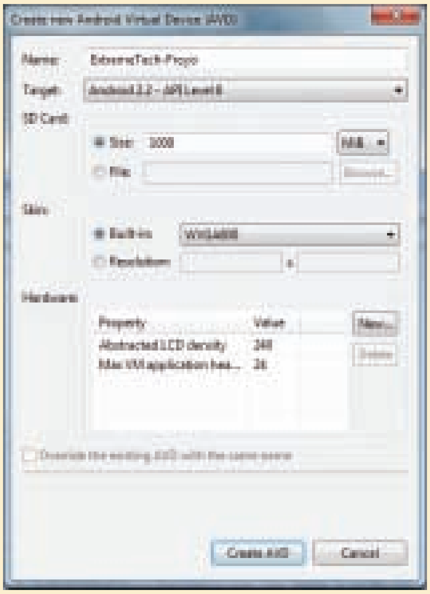
this example, create a 1GB SD Card. You can use larger sizes as well.
Skin: You can choose any pre-defined (built-in) or custom resolution that you want—it doesn't have to match the resolution or aspect ratio of a real phone screen. For this scenario, chosse WVGA800 (800x480), which is the screen resolution of a number of phones, such as the HTC Incredible. If you wanted to, you could set a custom resolution that simulated what an Android tablet might look like—perhaps with a screen resolution of 1024x600.
Hardware: These items will be selected automatically. You don't need to make any additions or changes here—unless you are a developer and you know what you are doing.

Now click the Create AVD button.

The AVD will be created lickety-split, and a window will pop up stating that the AVD has been created. Click OK to close the window. Note that once you create an AVD, its settings can't be changed. If want to make any changes, you'll need to create a brand new AVD. The AVD you just created should now appear in the list of Virtual Devices.

To launch the AVD, select it and then click the Start button. A Launch Options window opens, which allows you to Scale the display and Wipe user data. You might be tempted to choose the Scale display option--but don't, because if you do, the resulting window will be the physical size of a phone screen (about only three inches tall). Don't select the



caption

Wipe user data option either (though we will be using this option later when we set up the Android 1.6 AVD that includes the Android Market). Just click the Launch button to start the AVD.

It might take a couple of minutes for the AVD to launch; so be patient. But soon enough, the AVD will appear on your screen. The AVD is also referred to as an emulator because it "emulates" an Android device.

You'll see a window on the left that is the device's display, and a window on the right that houses the device's navigation buttons and virtual keyboard. In most cases you seldom have to use the onscreen navigation buttons or keyboard—your system's real mouse and keyboard should do the trick.

Here are some useful mappings to help you navigate in the AVD using your system's keyboard:

Home	Go to the Home Screen
F2	Menu Button
ESC	Back Button
F6	Toggle Trackball Mode
Ctrl+F5	Increase Volume
Ctrl+F6	Decrease Volume
Ctrl+F11	Switch Between Portrait and Landscape Mode
Alt+Enter	Toggle Full-Screen Mode

Before you start exploring, one of the first things you might want to do is confirm that the AVD has Internet access. Look in the status bar (running across the top of the AVD's display screen) and determine if you see an icon that has four bars with the first two bars lit up. If so, then the AVD at least thinks it has Internet access. You can confirm that the AVD's Internet access works by clicking on the Web browser icon in the bottom panel of the home screen—this is the icon that looks like a globe—and seeing if it connects to the Internet.

The AVD is simply piggybacking off of your system's Internet connection. While experimenting with the Android SDK on a number of different systems, We encountered a few situations where an AVD's Internet connection did not work. In all of these instances, the system was connected to the Internet via an Ethernet cable; but the system also had a Wi-Fi radio that wasn't connected to an access point. Connecting the system's Wi-Fi radio to an access point and restarting the AVD resolved the Internet access issue for me.

If you click the Launcher icon (the icon that is made up of 4x4 squares), you'll see the apps that come preinstalled in the AVD. You're likely going to be disappointed—there aren't many apps here at all. But fret not; in the next section we explain how to install Android apps onto your AVD.

But before we discuss installing apps, a few words on quitting and starting an AVD. Quitting an AVD is as simple as just closing the window. Really. That's all there is to it.

As to starting an existing AVD, there are a number of different ways to this. One is to launch the Android SDK and AVD Manager (SDK Setup.exe) application, and then start the AVD from the Virtual Devices tab. But if you're going to be using the same AVD on a regular basis, here's how you can create a shortcut for it, so you can launch the AVD directly from your desktop:

1. Determine the exact name of the AVD you want the shortcut to launch. You can get this info either from the Android SDK and AVD Manager or from the physical location where the AVDs are saved, which is C:\Users\[username]\.android\avd\.
2. Create a new desktop shortcut for the emulator.exe file that resides in the ...\android-sdk-windows/tools folder.
3. Once you've created the shortcut, edit its properties and add these options to the end of the target text field: -avd [name-of-avd]. This isn't case-sensitive, so you can do it all in lowercase. For my example, the shortcut's full target text field reads as follows: c:\android-sdk-windows\tools\emulator.exe –avd extremetech-froyo

Now whenever you want to launch the AVD, all you have to do is double-click the shortcut.

Installing Apps From Independent Android Apps Stores

If you've followed these instructions up to this point, you've got an AVD running Android 2.2, with a few barely useful apps and no Android Market. So how are you going to get apps onto your AVD? By downloading them from one of the independent Android apps stores.

Using your computer's Web browser, navigate to one of the Android apps stores, such as Softonic, Handango, or GetJar). Find the apps you want to